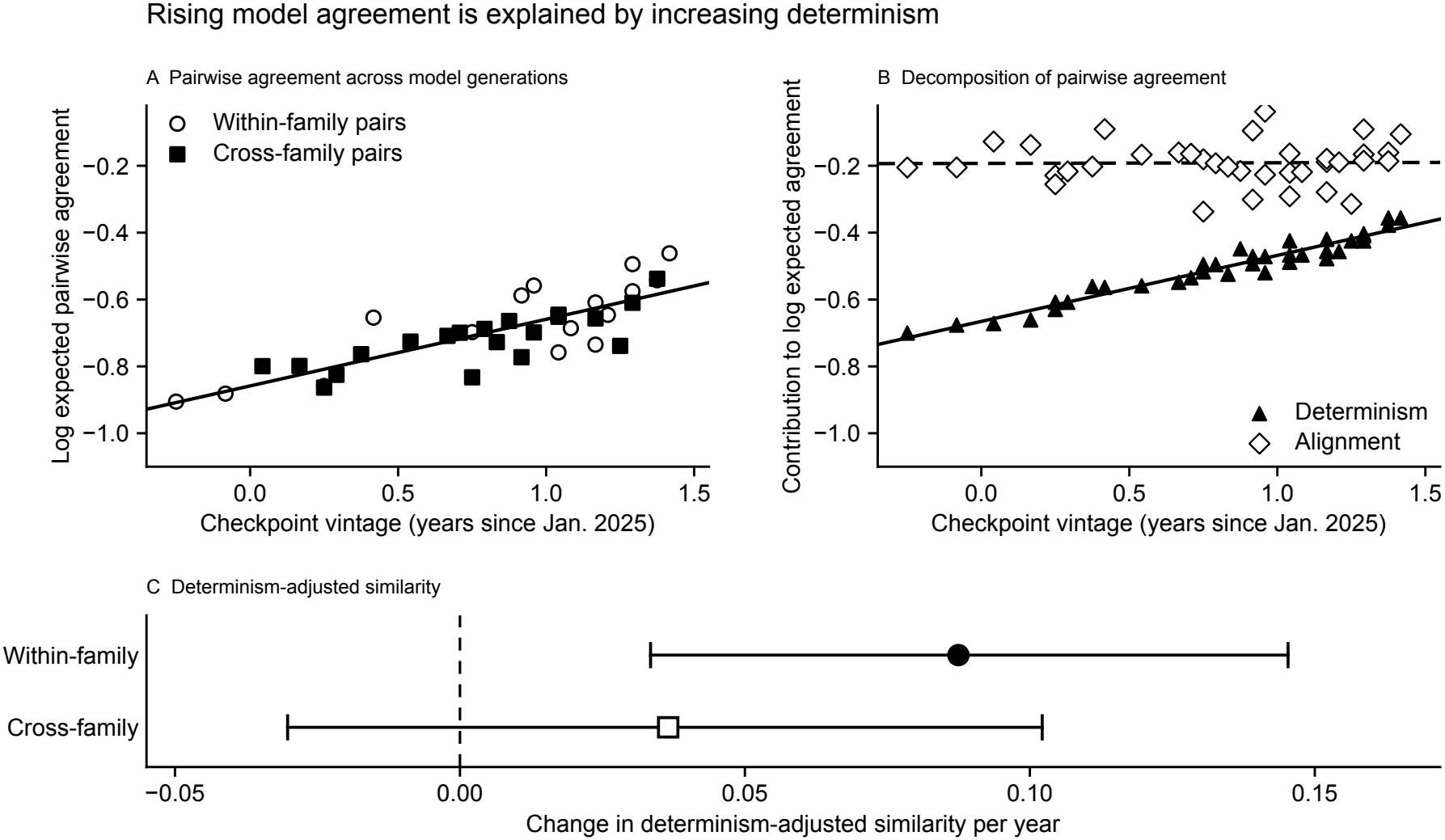Rising model agreement is explained by increasing determinism